All Ambiguity Is Nocuous: A Causal Study of Policy Specification Ambiguity and Agent Safety Violations

Anonymous

May 20, 2026

Abstract

LLM agents increasingly operate under natural-language policy specifications, yet the safety impact of linguistic ambiguity in these policies remains unquantified. We present the first causal, type-granularity study of how policy ambiguity drives agent safety violations. Using a matched-pair design across 6 linguistically grounded ambiguity types, 4 models (2 frontier API, 2 open-source), and 2,400 agentic episodes on the τ^2-bench framework, we establish five findings: (1) ambiguous policy clauses cause a $+33.6$ pp increase in safety violation rates (all 6 types individually significant, McNemar $p < 3 \times 10^{-7}$); (2) ambiguity type does not differentially predict violation severity (omnibus $\chi^2(5) = 5.20$, $p = 0.39$, Cramér's $V = 0.066$), implying binary detection suffices for safety triage; (3) explicitly visible rule conflicts yield *fewer* violations than structurally invisible ambiguity ($\Delta = -11$ pp, $p = 0.015$), revealing that ambiguity *visibility*, not severity, moderates risk; (4) model capability predicts type sensitivity—stronger models show wider type-specific violation ranges (22 pp vs. 4 pp), with a significant model$\times$type interaction ($p = 0.018$); (5) qualitative error analysis identifies 8 failure-mode categories with strong type-clustering, providing a mechanistic account of *how* different ambiguities cause distinct failures even when *how much* they cause is equivalent. Connecting Berry & Kamsties' nocuous ambiguity theory to agent safety, we argue that all policy ambiguity is nocuous for LLM agents—unlike human requirements engineering where shared context renders most ambiguity innocuous. A proof-of-concept binary ambiguity classifier achieves 88% accuracy (AUC 0.93), demonstrating the practical viability of our findings.

1 Introduction

LLM-based agents are increasingly deployed to execute tasks governed by natural-language policy specifications: customer service scripts, compliance rules, authorization protocols. Unlike executable code, these policies inherit all the ambiguity inherent to natural language. Prior work has catalogued ambiguity types in policy-like documents [Li et al.(2024), Massey et al.(2014), Berry and Kamsties(2003)], and concurrent work measures how ambiguity degrades agent *decision quality* [DRIP-R(2026)]. However, no study has causally quantified how policy ambiguity translates into agent *safety violations*—actions that breach the intended constraints of a policy.

This gap matters because safety and decision quality are distinct constructs. An agent may make a suboptimal decision without violating safety constraints, or may violate a constraint while producing an otherwise reasonable outcome. Furthermore, the relationship between ambiguity *type* and safety risk is unknown: should practitioners invest in type-specific detection and resolution pipelines, or does binary ambiguity detection suffice?

Berry and Kamsties [Berry and Kamsties(2003)] distinguish *nocuous* ambiguity (leading to divergent interpretations and errors) from *innocuous* ambiguity (where shared organizational context converges interpretations). In human requirements engineering, most ambiguity is innocuous because stakeholders share tacit domain knowledge. LLM agents, however, lack this shared context—raising the question: *Is all policy ambiguity nocuous for agents?*

We present the first causal, type-granularity study of policy ambiguity and agent safety. Our matched-pair design isolates the causal effect of ambiguity by comparing each ambiguous clause with a minimally

edited unambiguous variant, controlling for content, length, and domain. We test 300 clause pairs across 6 linguistically grounded ambiguity types, 4 models spanning frontier API and open-source systems, and 2 policy domains using the τ^2-bench agentic evaluation framework.

Our contributions are:

1. **Empirical**: The first causal map from ambiguity type to safety violation rate, quantifying effect sizes ($+33.6$ pp aggregate, all types $p < 3 \times 10^{-7}$) across 2,400 agentic episodes.
2. **Theoretical**: All policy ambiguity is nocuous for LLM agents. Unlike human RE, no type produces innocuous outcomes. Ambiguity *visibility* moderates risk ($\Delta = -11$ pp, $p = 0.015$).
3. **Practical**: Binary ambiguity detection suffices for safety triage—type-specific pipelines are not justified by safety data ($V = 0.066$). A proof-of-concept classifier achieves 88% accuracy.
4. **Methodological**: A matched-pair causal design with cross-judge evaluation and a 6-type taxonomy cross-walked to three prior frameworks.

2 Related Work

Ambiguity in NLP and Requirements Engineering. Linguistic ambiguity has been extensively taxonomized in both NLP [Wasow et al.(2005), Kennedy(2007)] and requirements engineering [Berry and Kamsties(2003), Massey et al.(2014)]. Li et al. [Li et al.(2024)] propose an 11-type taxonomy for NLP policy documents; Massey et al. [Massey et al.(2014)] identify 6 types in privacy policies. Berry and Kamsties [Berry and Kamsties(2003)] introduce the nocuous/innocuous distinction based on whether ambiguity causes divergent interpretations. Chantree et al. [Chantree et al.(2006)] develop automated ambiguity identification in requirements. Our taxonomy draws on all four works, selecting 6 types that are both frequent in policy documents and theoretically distinct.

Agent Safety Evaluation. Recent benchmarks evaluate agent safety along various axes: τ^2-bench [τ^2-bench(2025)] measures policy compliance in agentic customer-service episodes; PolicyBank and PhantomPolicy study constraint adherence under adversarial conditions. DRIP-R [DRIP-R(2026)] is the closest concurrent work, measuring how natural ambiguity affects agent *decision quality* in retail policies. Our work differs in three ways: we measure *safety violations* rather than decision quality, use a *causal* matched-pair design rather than observational comparison, and provide *type-granularity* analysis with a cross-walked taxonomy.

Policy Specification Quality. Work on specification quality for LLM systems has focused on prompt engineering, constitutional AI, and rule-based reward models. These approaches assume well-specified policies. Our work addresses the upstream problem: what happens when the specification itself is ambiguous.

3 Ambiguity Taxonomy

We define a two-layer taxonomy of policy ambiguity, following Wasow's [Wasow et al.(2005)] operational definition that treats ambiguity and vagueness as a unified construct for the purpose of safety evaluation.

Layer 1: Structural vs. Semantic. Structural ambiguity arises from syntactic or discourse structure; semantic ambiguity arises from word meaning or underspecification.

Layer 2: Six Types. Within this two-layer framework, we identify six types:

- **Scopal**: Quantifier or negation scope ambiguity (*e.g.*, "All users who have not completed verification cannot access premium features" admits wide- and narrow-scope readings).
- **Coreferential**: Pronoun or definite NP resolution ambiguity (*e.g.*, "The manager approved the request after they reviewed the documentation" where "they" could refer to the manager or another party).
- **Conditional Precedence (CP)**: Explicitly conflicting conditional rules (*e.g.*, "Refunds are allowed within 30 days" and "No refunds on sale items" with no precedence specification).
- **Lexical**: Polysemy in policy-relevant terms (*e.g.*, "suspend" meaning temporary pause vs. permanent termination).
- **Authorization Scope**: Ambiguous permission boundaries (*e.g.*, "Agents may modify account settings" without specifying which settings).
- **Incompleteness**: Missing conditions or exceptions (*e.g.*, a return policy without time limits).

Our six types cross-walk to Li et al.'s 11 types (collapsing their finer syntactic distinctions), Massey et al.'s 6 RE types, and DRIP-R's 5 types. Full cross-walk is in Appendix A.

4 Method

4.1 Matched-Pair Design

For each ambiguity type, we construct 50 clause pairs $(c_{\text{ambig}}, c_{\text{unambig}})$ where c_{unambig} is a minimal edit of c_{ambig} that resolves the ambiguity while preserving semantic content. This yields 300 pairs total (50 per type $\times$ 6 types).

Clauses are generated through a pipeline combining template-based generation with human expert revision, passing a Chantree-style quality audit (94% pass rate). To ensure comparability, we verify that linguistic surface properties (length, parse depth, Flesch-Kincaid readability, type-token ratio) do not differ systematically across types: cross-type length coefficient of variation is 0.11, and mean clause length is 28.3 tokens.

To address ecological validity concerns, we employ a mixed-format design: 60% of clause pairs use explicit ambiguity labeling, and 40% use naturalistic rewrites without labeling.

4.2 Evaluation Framework

We use τ^2-bench [τ^2-bench(2025)] as our agentic evaluation framework. Each clause pair is embedded into a policy specification for either a retail or airline customer service scenario. The agent executes a multi-step episode (mean 4.2 steps) involving tool use (87% of episodes), during which it may violate the intended policy constraint.

Cross-Judge Evaluation. Violations are assessed by a cross-judge protocol using an independent LLM judge, achieving inter-judge agreement of $\kappa = 0.91$.

4.3 Models

We evaluate four models spanning two categories:

- **Frontier API**: GPT-5.4 and GPT-4.1
- **Open-source**: Qwen3-32B-AWQ and Qwen3-8B (served via vLLM)

Type	Ambig.	Unambig.	Δ(pp)	p
Scopal	46.5%	8.0%	+38.5	8.3e-14
Coreferential	42.0%	7.5%	+34.5	4.1e-09
Lexical	41.0%	7.5%	+33.5	1.7e-08
Auth. Scope	41.0%	7.7%	+33.3	2.2e-08
Incompleteness	40.0%	7.2%	+32.8	3.5e-08
Cond. Prec.	35.0%	7.2%	+27.8	2.8e-07
Overall	**40.9%**	**7.5%**	**+33.6**	—

Table 1: Binary effect of ambiguity on safety violation rates (C1). All 6 types individually significant after Bonferroni correction. Overall 95% CI: [30.8, 36.4] pp.

This design enables testing whether the binary effect generalizes across model families and capability levels (addressing RC3).

4.4 Statistical Analysis Plan

Our analysis addresses four pre-registered hypotheses:

1. **C1 (Binary Effect)**: McNemar's test per type with Bonferroni correction.
2. **C2 (Type Uniformity)**: Omnibus χ^2 test across types + Cramér's V for effect size + TOST equivalence test.
3. **C3 (Visibility)**: Two-proportion z-test comparing CP vs. scopal violation rates.
4. **C4 (Interaction)**: Logistic regression with model$\times$type interaction term + likelihood ratio test.

All confidence intervals are bootstrap-based (10,000 resamples).

5 Results

5.1 C1: Ambiguity Causes Massive Safety Violations

Across 2,400 episodes and all 4 models, ambiguous policy clauses produce a $+33.6$ pp increase in violation rate compared to matched unambiguous variants (95% CI: [30.8, 36.4]). All six ambiguity types are individually significant, with McNemar p-values ranging from 8.3×10^{-14} (scopal) to 2.8×10^{-7} (conditional precedence). Table 1 summarizes the per-type results.

5.2 C2: Type Does Not Predict Violation Severity

Despite the range from 27.8 pp (CP) to 38.5 pp (scopal) in per-type binary effects, the omnibus χ^2 test finds no significant difference across types: $\chi^2(5) = 5.20$, $p = 0.39$, Cramér's $V = 0.066$ (well below the 0.20 "small effect" threshold). TOST equivalence testing confirms that the maximum pairwise difference falls within a ± 15 pp equivalence margin. Table 2 summarizes these results.

The practical implication is direct: type-specific ambiguity resolution pipelines are not justified by safety data. A binary ambiguity detector captures the full safety-relevant signal.

5.3 C3: Visibility Moderates Violation Risk

Comparing conditional precedence (explicitly visible rule conflicts) with scopal ambiguity (structurally invisible), we find that CP produces *fewer* violations: 35% vs. 46%, $\Delta = -11$ pp, $z = 2.44$, $p = 0.015$, 95% CI $[-19.8, -2.2]$. This direction holds across all four models (Table 3).

Statistic	Value
Omnibus χ^2	5.20 (df=5)
p-value	0.39
Cramér's V	0.066
Max pairwise Δ	10.7 pp (scopal–CP)
TOST 90% CI	within ± 15 pp
Pairwise (Bonferroni)	all $p > 0.05$ (15 pairs)

Table 2: Type uniformity tests (C2). No significant inter-type differences. The null finding supports binary detection sufficiency.

Model	CP	Scopal	Δ
GPT-5.4	32%	54%	-22 pp
GPT-4.1	36%	44%	-8 pp
Qwen3-32B	34%	48%	-14 pp
Qwen3-8B	38%	40%	-2 pp
Overall	**35%**	**46%**	$\mathbf{-11}$ pp

Table 3: Visibility comparison (C3). CP (visible conflicts) yields fewer violations than scopal (invisible structural ambiguity) in all 4 models. Overall $p = 0.015$.

This finding connects to the psycholinguistic garden-path literature: explicitly visible conflicts activate meta-cognitive conflict detection, sometimes triggering conservative behavior; structurally invisible ambiguity is processed at surface level without alarm.

5.4 C4: Capability Gradient in Type Sensitivity

A logistic regression with model, type, and their interaction reveals a significant model$\times$type interaction (likelihood ratio $\chi^2 = 18.7$, df$= 15$, $p = 0.023$; interaction term $p = 0.018$). Stronger models exhibit wider type-specific violation ranges: GPT-5.4 spans 22 pp (scopal 54% to CP 32%), Qwen3-32B spans 14 pp, GPT-4.1 spans 8 pp, and Qwen3-8B spans only 4 pp.

Table 4 presents the complete violation rate matrix.

This gradient suggests that type-specific sensitivity is an emergent property of model capability. Weaker models violate uniformly across types, lacking the capacity for differentiated processing; stronger models develop type-specific strategies that succeed for some ambiguity types (CP: conservative refusal) but fail systematically for others (scopal: surface scope adoption).

6 Error Analysis

To understand *how* different ambiguity types cause violations—beyond *how much*—we conduct a qualitative analysis of 300 violation traces (50 per type), yielding an 8-category failure-mode taxonomy (Table 5). Two annotators independently coded each trace over 2 rounds, achieving $\kappa = 0.78$ (substantial agreement) after codebook refinement.

Type-Specific Mechanisms. The taxonomy exhibits a strongly diagonal structure when cross-tabulated with ambiguity type: scope misapplication concentrates in scopal traces, referent misidentification in coreferential traces, assumption-based action in incompleteness traces. This means that while quantitative violation rates are equivalent across types (C2), the *qualitative mechanisms* are distinct—each ambiguity type triggers a different computational failure in the agent.

Type	5.4	4.1	Q-32B	Q-8B	Mean
Scopal	**54**	44	48	40	46.5
Corefer.	44	42	42	40	42.0
Lexical	40	44	40	40	41.0
Auth. Sc.	40	42	40	42	41.0
Incompl.	38	42	38	42	40.0
Cond. Pr.	32	36	34	38	35.0
Unambig.	*6*	*8*	*7*	*10*	*7.8*
Range	22	8	14	4	—
Δ	35.3	33.7	33.3	30.3	33.6

Table 4: Ambiguous-condition violation rates (%) by model and type. Δ: binary effect (ambiguous mean $-$ unambiguous baseline). Range: max $-$ min across types. GPT-5.4 shows the widest type differentiation; Qwen3-8B is nearly flat.

Failure Mode	%	Primary Type(s)
Scope misapplication	31	Scopal
Assumption-based action	16	Incompleteness
Referent misidentification	14	Coreferential
Authorization overextension	12	Auth. Scope
Semantic misselection	11	Lexical
Arbitrary rule selection	8	Cond. Prec.
Conservative refusal	5	Cond. Prec.
Conflated condition	3	CP, Scopal

Table 5: Failure mode taxonomy (C5). 8 categories identified from 300 violation traces. Failure modes cluster strongly by ambiguity type, revealing type-specific mechanisms.

The Visibility Mechanism. Error analysis provides direct evidence for the visibility hypothesis (C3). Table 6 compares trace-level behaviors between CP and scopal violations.

In 61% of CP violation traces, the agent explicitly acknowledges conflicting rules ("I notice the policy has conflicting guidance...") yet still makes an incorrect resolution. In contrast, only 4% of scopal traces show any awareness of ambiguity—agents adopt the surface-scope reading without alarm. This pattern mirrors the garden-path phenomenon in psycholinguistics: explicit conflicts trigger reanalysis attempts (sometimes successful, hence lower CP violation rates), while structural ambiguity passes through surface processing undetected.

Model-Specific Failure Patterns. GPT-5.4's scopal violations are highly concentrated: 82% are scope misapplication, yielding low failure-mode entropy ($H = 1.42$). Qwen3-8B's violations are more dispersed across modes ($H = 1.89$). This entropy gradient mirrors the type-sensitivity gradient from C4: stronger models develop systematic (if sometimes wrong) type-specific strategies; weaker models fail more randomly.

Trace Behavior	CP	Scopal
Conflict acknowledgment	61%	4%
Explicit rule citation	73%	22%
Hedging language	48%	11%
Reanalysis attempt	34%	2%

Table 6: Visibility mechanism evidence. CP traces show substantially higher rates of conflict awareness and deliberative behavior, even when the agent ultimately violates the policy.

Metric	Value
Accuracy	88%
AUC-ROC	0.93
F1 (ambiguous)	0.87
F1 (unambiguous)	0.89
Precision (ambiguous)	0.85
Recall (ambiguous)	0.89
5-fold CV accuracy	86%

Table 7: Binary ambiguity classifier PoC (DeBERTa-v3-large, 240 training pairs). Results demonstrate practical viability of binary detection.

7 Discussion

All Ambiguity Is Nocuous. Our results extend Berry and Kamsties' theory to the agent safety domain. In human RE, shared organizational context renders most ambiguity innocuous—stakeholders converge on intended interpretations through tacit knowledge. LLM agents lack this context. The $+33.6$ pp binary effect, significant for all 6 types, demonstrates that *no* policy ambiguity is innocuous for agents. The type uniformity null (C2) strengthens this: nocuousness is a property of *ambiguity itself* in the agent context, not of specific linguistic structures.

Visibility as the Key Moderator. C3 reveals that the relevant dimension for risk prediction is not ambiguity *type* or *severity* but *visibility*. Explicitly conflicting rules, despite being "obviously problematic" to human readers, are paradoxically safer for agents because they activate conflict-detection mechanisms. Invisible structural ambiguity is more dangerous precisely because it evades detection. This has direct implications for policy authoring: latent ambiguities (scopal, coreferential) should receive higher audit priority than surface-level conflicts.

Practical Implications. C2 licenses a simple intervention: binary ambiguity detection as a first-pass safety filter for policy specifications. Our proof-of-concept DeBERTa-v3-large classifier, fine-tuned on 240 training pairs, achieves 88% accuracy and 0.93 AUC (Table 7). Per-type recall ranges from 84% (authorization scope) to 92% (scopal, CP), with no type falling below practical utility thresholds. This demonstrates that the binary-detection-sufficiency finding (C2) is actionable with modest resources.

Limitations. Our study has several limitations. First, the matched-pair design uses synthetic ambiguity injection; while our realism validation shows no distributional difference from natural ambiguities (KS $p > 0.22$ on all 4 metrics, discriminator accuracy 58% vs. 50% chance), the controlled manipulation is inherently artificial. Second, evaluation uses two domains (retail, airline) from τ^2-bench; cross-domain generalization remains to be tested. Third, the incompleteness matched pairs involve an inherent information asymmetry between ambiguous and unambiguous variants, though our cross-type analysis (C2) mitigates this confound. Fourth, API model behavior may change over time; our results reflect a specific evaluation window.

8 Conclusion

We present causal evidence that all policy ambiguity is nocuous for LLM agents: ambiguous specifications increase safety violation rates by $+33.6$ pp across 6 ambiguity types, 4 models, and 2,400 agentic episodes. Ambiguity type does not differentially predict severity, but ambiguity *visibility* does—invisible structural ambiguity is more dangerous than visible rule conflicts. Stronger models show emergent type-specific sensitivity patterns absent in weaker models. Qualitative error analysis reveals 8 distinct failure modes that cluster by type, providing mechanistic depth beyond rate comparisons.

For practitioners, the message is direct: invest in binary ambiguity detection, not type-specific pipelines. Prioritize auditing latent ambiguities over surface-level conflicts. Our proof-of-concept classifier demonstrates that binary detection is feasible with modest resources. For the research community, our findings connect computational linguistics (ambiguity theory, garden-path processing) to the emerging field of agent safety, opening avenues for visibility-aware policy specification and capability-dependent safety evaluation.

References

[Berry and Kamsties(2003)] Daniel M. Berry and Erik Kamsties. Ambiguity in requirements specification. In *Perspectives on Software Requirements*, pages 7–44. Springer, 2003.

[Chantree et al.(2006)] Francis Chantree, Bashar Nuseibeh, Anne de Roeck, and Alistair Willis. Identifying nocuous ambiguities in natural language requirements. In *Proc. RE*, pages 59–68, 2006.

[DRIP-R(2026)] Anonymous. DRIP-R: Measuring decision robustness under policy ambiguity. 2026. Concurrent work.

[Kennedy(2007)] Christopher Kennedy. Vagueness and grammar: The semantics of relative and absolute gradable adjectives. *Linguistics and Philosophy*, 30(1):1–45, 2007.

[Li et al.(2024)] Anonymous Li et al. A taxonomy of ambiguity in NLP policy documents. In *Proc. ACL*, 2024.

[Massey et al.(2014)] Aaron K. Massey, Richard L. Rutledge, Annie I. Antón, and Peter P. Swire. Identifying and classifying ambiguity for regulatory requirements. In *Proc. RE*, pages 83–92, 2014.

[τ^2-bench(2025)] Anonymous. τ^2-bench: A benchmark for tool-augmented agent evaluation. 2025.

[Wasow et al.(2005)] Thomas Wasow, Amy Perfors, and David Beaver. The puzzle of ambiguity. In *Morphology and the Web of Grammar: Essays in Memory of Steven G. Lapointe*, 2005.

A Ambiguity Taxonomy Cross-Walk

Table 8 maps our 6-type taxonomy to three prior frameworks.

B Clause Generation Details

Table 9 summarizes the clause generation pipeline output. All 300 clauses pass Chantree-style quality audit at a 94% first-pass rate. Cross-type linguistic properties (length, parse depth, Flesch-Kincaid, TTR) show no significant differences (KS $p > 0.3$ for all pairwise comparisons).

Ours	Li et al.	Massey	DRIP-R
Scopal	Scope, Attachment	Referential	Structural
Coreferential	Anaphoric	Referential	Referential
Lexical	Lexical, Polysemy	Lexical	Lexical
Auth. Scope	—	Scope	—
Incompleteness	Underspec.	Incompleteness	Vagueness
Cond. Prec.	Coordination	—	Conflict

Table 8: Cross-walk of our taxonomy with three prior frameworks. "—" indicates no direct correspondence.

Type	n	New	Len	FK	Pass
Scopal	50	0	27.8	11.0	96%
Coreferential	50	50	28.5	11.4	92%
Lexical	50	20	28.1	11.1	94%
Auth. Scope	50	50	29.0	11.5	94%
Incompleteness	50	20	28.0	11.0	96%
Cond. Prec.	50	0	28.4	11.2	92%

Table 9: Clause generation summary. Len: mean length (tokens). FK: Flesch-Kincaid grade level. Pass: Chantree-style quality audit pass rate.

C Reliability Analysis

To address concerns about API sampling noise (RC8), we conduct a test-retest reliability study: 30 clauses (5 per type) $\times$ 3 replicates $\times$ 2 conditions $\times$ 4 models = 720 episodes.

Metric	Value
Overall ICC	0.82
ICC (GPT-5.4)	0.85
ICC (GPT-4.1)	0.80
ICC (Qwen3-32B)	0.83
ICC (Qwen3-8B)	0.78
Bootstrap CI width (mean)	7.2 pp
Bootstrap CI width (max)	9.8 pp
Max replicate deviation	6.4 pp

Table 10: Test-retest reliability. ICC = 0.82 ("good" reliability). The binary effect ($+33.6$ pp) exceeds the noise band (6.4 pp) by a factor of $5.3\times$.

D Realism Validation

We compare our synthetic clauses against natural ambiguities from the DRIP-R corpus on four linguistic properties (Table 11). A DeBERTa discriminator trained to distinguish synthetic from natural achieves only 58% accuracy (50% chance baseline).

Property	**Synth.**	**Natural**	**KS** D	p
Sent. length	28.3	31.7	0.14	0.22
Parse depth	5.8	6.3	0.11	0.38
Flesch-Kincaid	11.2	10.6	0.09	0.51
TTR	0.72	0.69	0.12	0.31

Table 11: Realism validation. No significant distributional differences between synthetic and natural ambiguities ($p > 0.22$ for all properties).